

(BOP) at a whopping 5 trillion dollars! The next level up in the pyramid are a section of 1.4 billion people who represent a consumer market worth 12.5 trillion dollars, so there's much more money in focussing on richer folks (as Apple will gladly tell you), but there's also much more competition. The BOP's 5 trillion market is virtually untouched by conglomerates and technology, and is mostly rural populations. But that also raises some challenges.

SPOTLIGHT ON INDIA

As we all know, our country is unlike any other on the planet, which makes things super complex, which is a good thing if you're looking to make a name for yourself. Complex problems need a multitude of solutions working together, and by default, make monopolies even harder to achieve. No single corporate giant has the skills or abilities or solutions to satiate all of India's needs and thus form a monopoly. Apple and Samsung and the likes might be able to rule the smartphone world in America and Europe, but in India, smaller brands are able to compete in a very price conscious market.

We have a vast land area, split by multiple languages and cultures, all with varying needs, which makes a one stop solution impossible. A simple example is needing to have multiple software keyboards for all of our languages. English and Hindi will only get you so far, and if we want the remaining 800 million people to come online, we're going to first need to pop up a keyboard that they can understand in order to type in something simple as a search term. We're going to need operating systems that are easy to use and, in their language. And of course, we're going to need content and services for them to be able to access (in their native language), in order to make it worth their while – they are expected to pay to have smartphones and net connections after all.

INTERESTING PROJECTS

Haptik

(<https://dgit.in/haptik>)

Haptik is a chatbot that was built for businesses, but with one big difference, it uses a conversational AI approach, so that people can converse naturally. We tried out the bot on their site and it seemed pretty much like the average well-programmed bot that we come across online. Very to the point and merely gives you options to select from. Trying to converse with it leads nowhere. However, they must be doing an awful lot right, because they've been acquired by Reliance Jio, presumably because Jio wants to target the next billion with conversational AI. For simple queries the chatbot interface should work perfectly well for new users.

We're not going to dwell too long on the specific problems India faces with technology, or look at who is doing what to solve those problems, because our August issue, which traditionally is an India special, is going to do exactly that. So we're content just scratching the surface here and leaving you waiting for more in the next month's issue.

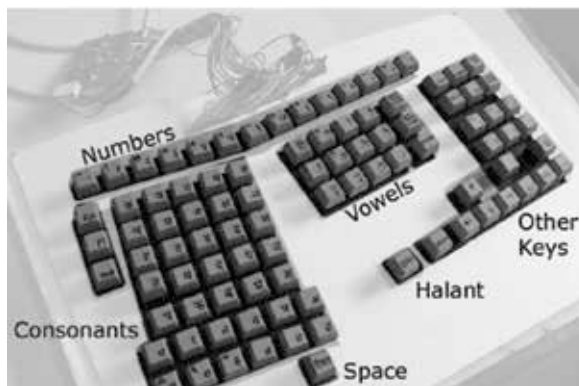
ARROGANCE

The biggest problem with the way we often look at the next billion is our own arrogance. Instead of asking them what they need, companies tend to assume what that is, and tailor solutions accordingly. An example is the whole Facebook Free Basics saga. While defending net neutrality is a noble cause, and perhaps was the right thing to do, no one really asked the next billion what they wanted, or what was stopping them from accessing the Internet. If it was indeed merely a cost issue, then perhaps a more neutral form of free basics was what was needed at the time.

However, the flip side of the coin is that Facebook was just as guilty of being arrogant and assuming they could play gatekeeper and decide what "poor" people should and shouldn't get access to. It's like throwing out scraps of substand-

ard net access, and then demanding that people be grateful for said scraps.

It's arrogant to think that the next billion are just aching to get on to the net the way it is structured right now, and that by changing nothing, and merely giving them free access to the internet, all of their problems will be solved. Access certainly is a problem,



The Barakhadi 1 Keyboard based on the Varnamala.

but it most definitely isn't all there is to the story.

Heck, it's probably arrogant of us to assume that people want or need vernacular keyboards, as we did earlier in this very article. Given that the lowest-end smartphone users in India, use a lot of voice search in vernacular languages. Of course it stands to reason that they're more comfortable talking to their phones as they would people, because that's may be the only way they know how to ask questions. Literacy is still a problem here, and keyboards might actually intimidate folk and make them not want to use them at all. A big mic button